USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 19 LAB - Fox Hunt
Lab Packet
Fall 2026

Names: _____ _____ **Section:** _____

Learning Objective 3.5: I can implement beam steering on the ADALM-PHASER and verify the steered pattern against theory.

One HB100 in the room is on, inside one of five numbered boxes placed inside a keep-out circle marked on the floor. You and a partner find the live box from bearings taken on your two kits from anywhere outside that circle. This packet is the L19 Lab, in three parts, and your team turns in one copy, complete, before Lesson 20. Both partners work **Part 1** before the lab period: the error budget, a plan for where to stand, and the triangulation script, tested on a supplied set of bearings. You need all three at the bench. **Part 2** is your team's measurements, recorded in class. **Part 3**, after the lab, turns the measurements into a fix and compares it with the box the instructor reveals.

Grading. One submission per team, counted toward both partners' engagement grades and graded as a genuine, documented attempt.

The array is the PHASER's: $N = 8$ elements, E1 to E8, on a $d = 14$ mm pitch. Use $c = 3.00 \times 10^8$ m/s and the HB100's nominal 10.525 GHz. Room angles are measured counterclockwise from the room's $+x$ axis, seen from above, and the origin is the center mark of the keep-out circle.

Part 1: Pre-Lab (both partners, before the lab period)

1. **The bearing error budget.** Est. Angle is the steer angle of the largest sample in the beam sweep, and the sweep steps the progressive phase one ADAR1000 LSB, $360^\circ/2^7 = 2.8125^\circ$, at a time.

   (a) Find kd in degrees, the sweep's step in $\sin\theta$, and the step in angle at each Est. Angle in the table. Then give the largest error the grid alone can cause at each angle.

Est. Angle	0°	30°	45°
Grid step			
Largest grid error			

   (b) You will aim each kit's broadside at the circle's center mark by eye, which holds the heading to about 1.5°. Take 1.0° for multipath and 0.2° for noise. Combine these with the grid term at 30° to find the bearing uncertainty σ_β. Then repeat with the heading set carefully to 1.0°.

σ_β (heading 1.5°) = ____ σ_β (heading 1.0°) = ____

(c) A teammate sets a kit on the floor, 0.75 m below the boxes, and 3 m from a box that is 30° off broadside in azimuth. What does Est. Angle read, and which way does the error always pull?

2. Where to stand. The keep-out circle has a radius of 1.5 m, and you may take bearings from anywhere outside it. Your room's radius may differ; the instructor gives the actual value at the start of the period. Take $\sigma_\beta = 2^\circ$ in this question.

(a) Find the cross-range error of one bearing line at 3, 4, and 5 m from the source.

Range	3 m	4 m	5 m
Cross-range error			

(b) Two kits 4 m from the source see it along bearings that cross at angle γ. Find the RMS error of the two-bearing fix at each crossing angle.

γ	90°	30°	15°
RMS fix error			

(c) Every box must stay within $\pm 30^\circ$ of each kit's broadside. A kit stands a distance D from the center mark with its broadside aimed at the center. Find the smallest D for which the whole circle stays within $\pm 30^\circ$ of broadside.

D_{min} = ____

(d) Sketch the circle and mark where your team plans to put its two kits and the second position each kit will move to, so that the team takes four bearings. Choose positions outside the circle that keep every box within $\pm 30^\circ$ of broadside and give steep crossings for a box anywhere in the circle. Label each position with its distance from the center and its direction in room angles.

(e) The five boxes are at least 0.75 m apart, and your team names the box nearest its fix. A fix closer to a wrong box than to the right one has to miss by at least half the spacing. Express half the spacing in units of the per-axis error, 0.14 m, of a 90° crossing, and say in one sentence whether naming the wrong box should be common.

3. **The triangulation script.** Write a Python function that takes the positions $\mathbf{p}_i$ of the kits and their room bearings β_i and returns the least-squares fix and each bearing's residual. Lesson 19's "Many Bearings: Least Squares" gives the normal equations. Use numpy; do not use MATLAB. Test it on this practice set from an earlier class, in which each kit stood outside the keep-out circle, aimed its broadside at the center mark, and read positive Est. Angle counterclockwise:

Bearing	**Position (m)**	**Heading** h	**Est. Angle**	**Room bearing** β
1	$(-3.0,\ -1.8)$	31.0°	$+1.4^\circ$	
2	$(-1.2,\ -3.3)$	70.0°	-6.7°	
3	$(1.2,\ -3.3)$	110.0°	-12.0°	
4	$(3.0,\ -1.8)$	149.0°	-3.5°	

Boxes: **1** $(-0.9,\ 0.6)$, **2** $(-0.2,\ -0.8)$, **3** $(0.6,\ 0.4)$, **4** $(0.0,\ 1.1)$, **5** $(0.9,\ -0.5)$ m

(a) Fill in the room bearings, run your function on all four, and record the fix and the four residuals. Attach your code.

	Fix (m)	e_1	e_2	e_3	e_4
All four					

(b) Each bearing is about 4 m from the fix, so a good bearing should leave a residual near $4\tan(2^\circ) = 0.14$ m. Which bearing is the outlier? Drop it, solve again with the other three, and record the new fix and residuals.

	Fix (m)	**Residuals of the three kept**		
Outlier dropped				

(c) Name the box from each fix, and give each fix's distance to that box.

Part 2: Measurements (your team, in class)

Follow the procedure on the L19 Lab page in read mode. Bring each kit up as in the L18 Lab, with your own HB100 at boresight for **Find HB100** and **Calibrate**. Load **1 Steering Angle**, set **Rx Gain (dB)** as the page says, and press **Start**. Your own HB100 stays on only for the reference shot. Then switch it off, disconnect its power, and, once the instructor starts the hunt, press **Find HB100** again so the kit tunes to the fox. Every HB100 runs at a slightly different frequency, and the receiver sees only a few megahertz around the one it is tuned to, so a kit still tuned to your own HB100 does not see the fox at all. Keep the calibration from bring-up. An HB100 radiates mainly forward, so a kit behind or well off the side of the live box may see a weak tone or none; if you cannot find the tone from a position, move.

Keep-out radius = ____

Own HB100, kit A = ____ kit B = ____

Fox, after re-tune = ____

4. **Reference shot.** Before the hunt, each partner places their own HB100 2 m in front of their kit and about 20° to the left of broadside, as seen from behind the array looking out along broadside. Record Est. Angle. If it reads positive, positive Est. Angle is counterclockwise on your kit and $s = +1$; if negative, $s = -1$.

Kit	Est. Angle, source left	s
A		
B		

5. **The hunt.** At each position outside the keep-out circle, aim the kit's broadside at the center mark, measure the position from the center mark along the room's taped axes, and read Est. Angle on ten consecutive sweeps from the **Tracking** tab. Record the mean, the spread (largest minus smallest), and the room bearing $\beta = h + s\,\bar{\theta}_{\text{est}}$, where h is the direction from the kit to the center.

#	Kit	Position (m)	h	Mean Est.	Spread	Peak Gain	β
1	A						
2	B						
3	A						
4	B						

6. **The call.** Run your Part 1 script on the four bearings in class, before the reveal. Record the fix, the largest residual, the box you name, and, after the instructor opens the boxes, the box that was live.

Fix (m)	Largest residual	Box named	Live box

Part 3: Writeup (your team, after the lab)

7. **Reconcile the fix.** The instructor posts the live box's measured position after the period. Answer each part in three or four sentences, using your numbers from Part 2 and your predictions from Part 1.

   (a) Give the distance from your fix to the live box, and compare it with the RMS fix error you predicted for your geometry in Question 2. Is the miss consistent with your budget?

   (b) Use the live box's position to find the true bearing from each of your four positions, and the error of each measured bearing. Which term of your error budget best explains the largest one?

   (c) If your residuals flagged an outlier, say whether dropping it moved the fix toward the live box or away from it, and what in the room you think caused it. If no residual stood out, say what that tells you about the four bearings.

8. **Changing the geometry.** Run your script on your own bearings again, using only bearings 1 and 2, then only 3 and 4, then only 1 and 3.

   (a) Record the three two-bearing fixes and the crossing angle of each pair. Which pair landed closest to the live box, and is that the pair with the steepest crossing?

   (b) In two or three sentences, explain why a two-bearing fix can never reveal its own bad bearing, and why your team took four.